SHREYASH V. THORAT

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EDUCATION

University of Southern California Los Angeles, CA.
Master of Science in Mechanical Engineering August 2025-May 2027

- Courses: AME504: Mechatronics, CSCI 445: Intro to Robotics, AME 547: Manufacturing Automation, AME 525: Linear Algebra. Overall **GPA: 3.6**

K. J. Somaiya College of Engineering Mumbai, India
Bachelors of Technology in Mechanical Engineering July 2020-January 2024

- Minors' degree in Artificial Intelligence & Machine Learning

RELEVANT SKILLS

- Programming and Platforms: Python, C++, Linux, MATLAB, Simulink, ROS2, Mujoco, SolidWorks, ANSYS, KiCAD
- Software Development and Libraries: Bash, Git, Docker, Data Structures and Algorithms
- Robotics: Kinematics and Dynamics, Sensor Fusion and Calibration, Control Systems, Motion Planning.

EXPERIENCE

Laboratory for Autonomous Systems in Exploration and Robotics (LASER) Los Angeles, CA
Graduate Student Researcher Nov 2025-Present

- Implementing a manipulator on a microgravity free-flyer air-bearing robot; developing an experimental setup and optimized gripper designs to characterize grasping strategies and autonomously identify optimal grasp candidates for in-space manipulation.

Space Engineering Research Center: Information Sciences Institute (USC) Los Angeles, CA
Graduate Student Researcher Sept 2025-Jan 2026

- Collaborating with hardware and propulsion team on the Leapfrog Lunar Lander prototype, analyzing test results, and proposing design iterations to improve landing stability

Greendzine Technologies Pvt Ltd Bangalore, India
Mechanical Design Engineer January 2024-May 2024

- Engineered and prototyped mechanical components for warehouse intra-logistics electric vehicles (100,000+ sq. ft. facilities), applying DFMA principles to improve manufacturability and reduce assembly time

Orion Racing India | Formula Student Team Mumbai, India
Suspension & Steering Subsystem Lead Feb 2021-Dec 2023

- Led a 12-member subsystem team through design reviews, sprint planning and working with departments to validate integration points, enhance design and resolve cross-system mechanical issues

ACADEMIC PROJECTS

Autonomous Mobile Robot: Localization, Planning & Control Los Angeles, CA
University of Southern California September 2025 - December 2025

- Implemented a **Particle Filter** localizer on a known 2D map using LIDAR scan matching and low-variance resampling, achieving pose convergence within a few exploratory motions as validated in Gazebo & Rviz.
- Built an **RRT motion planner** with obstacle-padded collision checking(with a **reactive safety layer** monitoring forward LIDAR returns), shortcut smoothing, and waypoint densification, consistently generating collision-free paths to goal in simulation
- Deployed a **dual-axis PID controller** for heading and velocity regulation, publishing to ROS topics within a closed sense–plan–act loop that corrected for odometry drift in real time via continuous LIDAR feedback

5-DOF Package Handling Robotic Arm for Pick and Place Operations Mumbai, India
Passion Project September 2024 - January 2025

- Designed a high-fidelity 5-DOF manipulator (700mm reach, 3kg payload) in SolidWorks, performing Inverse Kinematics (IK) analysis to define workspace limits and singularity avoidance.
- Integrated a ROS 2 control stack with MoveIt, utilizing motion planning to execute efficient pick & place trajectories in Gazebo.